

Section 2.4 Matrix Inverses

Exercise 2.4.2

Find the inverse of each of the following matrices.

e.
$$\begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$

i.
$$\begin{bmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

Exercise 2.4.3

In each case, solve the systems of equations by finding the inverse of the coefficient matrix.

d.

$$\begin{aligned}x + 4y + 2z &= 1 \\2x + 3y + 3z &= -1 \\4x + y + 4z &= 0\end{aligned}$$

Exercise 2.4.5

Find A when:

d. $(I - 2A^T)^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

f. $(\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} A)^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$

h. $(A^{-1} - 2I)^T = -2 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Exercise 2.4.7

Given $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 4 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ and $\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 0 \\ -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, express the variables x_1 , x_2 , and x_3 in terms of z_1 , z_2 , and z_3 .

Exercise 2.4.17

If $c \neq 0$, find the inverse of $\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 2 \\ 0 & 2 & c \end{bmatrix}$ in terms of c .

Section 2.5 Elementary Matrices

Exercise 2.5.2

In each case find an elementary matrix E such that $B = EA$.

a. $A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix}$

e. $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, B = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$

Exercise 2.5.3

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$.

- a.** Find elementary matrices E_1 and E_2 such that $C = E_2E_1A$.

Exercise 2.5.6

In each case find an invertible matrix U such that $UA = R$ is in reduced row-echelon form, and express U as a product of elementary matrices.

b. $A = \begin{bmatrix} 1 & 2 & 1 \\ 5 & 12 & -1 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 3 & 1 & 1 & 2 \\ 1 & -3 & 3 & 2 \end{bmatrix}$

Exercise 2.5.8

In each case factor A as a product of elementary matrices.

b. $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$

d. $A = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 4 \\ -2 & 2 & 15 \end{bmatrix}$